VectorQuay
INDIGO SIGNAL

TECHNICAL REQUIREMENTS DOCUMENT

Phase 0 Technical Requirements

Research, feasibility, enforcement policy, and delivery requirements for the first go/no-go decision.

VERSION	STATUS
v1.0	**In Review**
OWNER	THEME
Brad Malia	**Indigo Signal**

Executive Summary

Phase 0 technical work does not implement the trading product yet. It defines the research tasks, technical decision outputs, enforcement model, feasibility checks, and test planning required to determine whether VectorQuay should proceed into implementation.

Primary Output

A defensible go/no-go decision backed by exchange constraints, account-action policy, asset-universe rules, and documented technical assumptions.

1. Scope

- Translate the approved Phase 0 PRD into concrete technical research and decision outputs.
- Define the enforcement model for the Phase 0 business rules.
- Determine whether the product can operate meaningfully at approximately $50 starting capital on Coinbase spot.
- Produce the technical inputs required for Phase 1 architecture and implementation planning.

2. Out of Scope

- Building production trading logic.
- Implementing live order placement.
- Implementing paper trading.
- Implementing AI-driven signal generation.
- Implementing the GUI application.

3. Technical Objectives

- Validate Coinbase API and product constraints relevant to the approved business policy.
- Define a technical enforcement model for allowed, prohibited, and approval-gated account actions.
- Define the initial tradable universe policy in technically enforceable terms.
- Define the minimum documentation and traceability required for future implementation and testing.
- Produce a clear technical recommendation on whether to proceed, narrow scope, raise capital, or stop.

3A. Research Evidence Standard

- Coinbase official documentation and API references must be treated as the primary source for exchange capabilities and constraints.
- Where possible, technical conclusions should be supported by direct account or product inspection in read-only mode.
- Any assumption that is not directly verified must be documented explicitly as an assumption.
- Each major feasibility conclusion must record its source basis and confidence level.

3B. Research Execution Method

- Phase 0 research must be performed as a structured investigation, not informal note-taking.
- For exchange capabilities and constraints, use Coinbase official documentation as the primary source and verify against live read-only account or product inspection where possible.
- For account-specific viability questions, inspect the target Coinbase account in read-only mode once the implementation phase supports safe observation.
- For product decisions affected by costs, document fee assumptions, spread assumptions, and minimum-order assumptions explicitly.
- Each research topic must produce a written finding in a consistent format: question, conclusion, source, confidence, impact on v1, and any unresolved assumption.
- Research outputs must be stored in the repository so future sessions can resume without depending on chat history.

4. Research Workstreams

Coinbase Feasibility

- Confirm Coinbase Advanced Trade spot product support relevant to the target account type.
- Confirm usable order types for v1 assumptions.
- Confirm minimum order sizes, increments, and relevant size constraints.
- Estimate the effect of fees and spread at approximately $50 capital.
- Identify any API or account restrictions that affect automated spot trading.

Tradable Universe

- Define approved quote currencies for v1, starting with USD and USDC.
- Define liquidity, spread, and market-quality thresholds as numeric values or explicit threshold formulas.
- Define whether direct cross-asset pairs are allowed or whether routing must be normalized.
- Define whether USDC is treated as a routing-only asset or a strategic defensive allocation.
- Document candidate v1 assets that satisfy the above constraints.
- Document rejected assets and the reason each asset failed the v1 viability rules.
- Define the expected technical behavior when no asset satisfies the v1 tradable-universe rules.

Control Policy

- Define how bot-owned orders are tagged and identified.
- Define default classification for unknown orders.
- Define enforcement behavior when human activity is detected.
- Define protected-asset modes and how they map to trade validation.
- Define a technical permissions matrix aligned to the PRD policy matrix.
- Set a fail-safe provisional default for mixed human/bot activity until final implementation behavior is approved.
- Define the required order-ownership identifier characteristics: uniqueness, namespace/prefix, persistence in audit trails, and non-ambiguous ownership mapping.

Documentation and Traceability

- Define document outputs required to preserve project context across sessions.
- Define how decisions, assumptions, and unresolved risks are logged.
- Define traceability from PRD rule to technical enforcement and later test coverage.

5. Technical Deliverables

- Phase 0 technical requirements document.
- Technical permissions matrix with enforcement notes.
- Coinbase feasibility findings summary.
- Tradable-universe policy draft for v1.
- Approved and rejected asset evaluation list with reasons.
- Order ownership and mixed-control policy draft.
- Security and secrets handling baseline for future implementation phases.
- Documented assumptions and unresolved risk register.
- Go/no-go technical recommendation.

5A. Enforcement Architecture

- Phase 0 must define a pre-trade policy validation layer that checks permissions, protected assets, quote-currency rules, and candidate asset viability before a trade is considered executable.
- Phase 0 must define a risk validation layer separate from policy validation.
- Phase 0 must define a single execution boundary through which all order submission and cancellation actions must pass.
- Phase 0 must define a post-trade reconciliation boundary responsible for comparing expected and actual account state.
- Phase 0 must define the audit logging boundary that records all policy checks, risk checks, approvals, denials, and execution outcomes.

6. Technical Permissions Matrix

Action	Business Policy	Enforcement Intent	Audit Requirement
Read balances/orders/fills/products/market data	Allowed	Allow through read-only service boundaries.	Log request and source context at debug/audit level.
Place spot buy/sell orders	Allowed	Only through strategy and risk-checked execution path.	Log candidate trade, approval result, and submitted order metadata.
Take no trade action	Allowed	System must be able to reject all candidates and remain in current allocation or defensive cash-like allocation.	Log no-trade decision and rejection reasons.
Cancel bot-owned orders	Allowed	Require positive bot ownership marker before cancellation.	Log ownership proof and cancel reason.
Cancel human-owned orders	Human approval only	Default deny unless explicit operator approval	Log denial or approval path with actor identity.

Action	Business Policy	Enforcement Intent	Audit Requirement
		workflow exists.	
Trade protected assets	Mode dependent	Validate `no buy`, `no sell`, or `fully excluded` before order creation.	Log protected-asset check result on every candidate trade.
Add, modify, or remove bank accounts/payment methods	Prohibited	No supported code path or API call may exist for v1.	Log attempted requests as policy violations if encountered.
Bank deposits / bank withdrawals	Prohibited	No supported code path or API call may exist for v1.	Log attempted requests as policy violations if encountered.
External wallet transfers	Prohibited	No supported code path or API call may exist for v1.	Log attempted requests as policy violations if encountered.
Recurring buys / non-strategy conversions / profit sweeps	Prohibited	No supported code path may exist outside the approved strategy engine.	Log attempted requests as policy violations if encountered.
Unknown orders without bot marker	Treat as human-	Default deny for cancellation	Log unknown-

Action	Business Policy	Enforcement Intent	Audit Requirement
	owned	or modification until ownership is proven.	order detection and resulting control action.
Unexpected human trading activity	Fail-safe handling required	Default provisional behavior must be restricted mode or pause until implementation policy is finalized.	Log human-activity detection, chosen fail-safe path, and operator notification requirement.
Quote-currency violations	Prohibited	Reject trade candidates that do not use approved v1 quote currencies.	Log rejection with quote-currency policy reason.
Staking / Web3 / margin / derivatives	Prohibited	No supported code path or API call may exist for v1.	Log attempted requests as policy violations if encountered.

7. Technical Decision Outputs Required

- Is approximately $50 starting capital technically viable after likely fees, spread, and minimum-order constraints?
- Which v1 assets satisfy the tradable-universe rules?
- Which assets were evaluated and rejected, and why?
- Should v1 standardize routing on USD only or USD plus USDC?
- Should the system take no action or remain in cash-like allocation when no candidate asset passes viability filters?
- What is the final required default behavior when mixed human/bot activity is detected?
- What order metadata scheme will identify bot ownership reliably, and what identifier properties are mandatory?

8. Security and Secrets

- No production or user-specific secrets may be committed to the repository.
- Phase 0 must define the intended local secret-handling approach for development and future deployment.
- Phase 0 must define minimum log-redaction expectations for secrets and sensitive account identifiers.
- Phase 0 must preserve the bring-your-own-credentials model required for future resale.
- Only configuration templates may be committed to source control; real values must remain external.
- Phase 0 must define expected configuration separation for local, development, and future production-like environments.

9. Non-Functional Technical Requirements

- All technical findings must be versioned in the repository.
- All technical rules must be traceable back to the approved PRD.
- Decisions must be understandable without reliance on chat history.
- The output must support future resale, multi-user packaging, and secret isolation.

10. Entry and Exit Criteria

Entry Criteria

- Approved roadmap v4 exists.
- Approved Phase 0 PRD exists.

Exit Criteria

- Technical permissions matrix is complete.
- Feasibility findings are documented with source basis and recorded assumptions.
- Tradable-universe rules are documented with quantitative thresholds or formulas.
- Approved and rejected asset lists are documented with reasons.
- Mixed human/bot control policy is documented.
- A final default behavior for mixed human/bot activity is selected.
- Unresolved risks and assumptions are explicitly recorded.
- A final technical decision outcome is produced: proceed as planned, proceed with narrower scope, proceed with higher required capital, or stop.

11. Risks and Unknowns

Key Technical Risks

- The account size may be too small to justify automated trading after costs.
- Coinbase account/product restrictions may narrow the usable asset universe more than expected.
- Mixed human/bot control may create state ambiguity without strong ownership tagging.
- Future resale needs may require stricter configuration boundaries than initially assumed.

12. Recommendation

Proceed with Phase 0 technical research and policy definition. No implementation phase should begin until the technical findings either confirm feasibility or narrow the scope to a technically defensible v1.

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